Comparative LCA:s for Wood and Other Construction Methods

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Summary

This paper summarises and discusses the results from seven different comparative LCA studies that have compared wood frame construction buildings with one or more alternative construction techniques. Although the physical boundaries (functional units) of the studied structures as well as the system boundaries for LCA:s differ significantly between the studies, the conclusion is very clear; The wood frame structure performs better with respect to energy use and "Global Warming Potential" (GWP) than the alternatives according to all comparative studies. The important issue of LCA system boundaries is discussed and it is concluded that system expansion should be used to be able to include the effect of the energy recycling potential from used wood based products.

Keywords: Comparative LCA, GWP, CO_2, Wood frame construction, steel structure, concrete structure, energy recycling, feedstock energy, system expansion.

1. Overview of the study and studied reports

This paper presents an overview of seven comparative life cycle assessments of buildings for housing with wood (frame) structures, concrete structures and steel structures or combinations of the three materials. It should be noted that it has not been within the scope of this overview to scientifically examine the quoted reports, their methodology or their LCI data. However, were the comparison cases notably deviate from each other, this is pointed out.

The studies in the overview and the results of each of these in terms of energy use and green-house gas emissions (expressed as GWP, Global Warming Potential) are summarised very briefly. Further information is provided in [1].

1.1 Case 1: Environmental Assessment of Trähus 2001, Trätek [2]

This case compares a four storey wood frame residential house ("Trähus 2001") erected in 2001 for the Bo01 housing fair in Malmö, Sweden with another four storey residential house within Bo01 (for simplicity called the "Concrete house"). "Trähus 2001" has a typical wood frame structure and wood cladding, whereas the "Concrete house" has a structure of steel and prefab concrete with exterior non-structural wall panels of wood frame construction and stucco cladding. The exterior wall heat insulation (and thus U values without regarding windows) differ somewhat with 215 mm mineral wool for "Trähus 2001" and 250 mm for the "concrete house", even though the buildings were designed to the same energy use standards specific for Bo01.

The system boundaries for the LCA are essentially "cradle to grave", but excluding the usage phase which is assumed equal for the two buildings. Recyclable energy from the buildings is accounted for in both cases, primarily for the wood based components (recycled *material* is regarded in the material manufacturing phase). The "Concrete house" is then (for CO_2 emissions calculations) burdened with the CO_2 emissions from an equal amount of energy generated from oil. In the energy use comparison the recyclable energy has been deducted from the cradle to gate energy need.

The LCA results for "Trähus 2001" are: Cradle to gate energy use: 960 MJ/m^2; Recyclable energy:

1460 MJ/m^2; Total energy use excluding usage phase: 960-1490 = -530 MJ/m^2; GWP (CO_2-equivalents): 30 kg/m^2. The comparable results for the "Concrete house" are: Cradle to gate energy use: 2260 MJ/m^2; Recyclable energy: 490 MJ/m^2; Total energy use excluding usage phase: 2260-490 = 1770 MJ/m^2; GWP (CO_2-equivalents): 400 kg/m^2.

1.2 Case 2: Building Life Cycle Assessment: Residential Case Study, Athena Sustainable Material Institute [3]

This case compares three alternative designs of a 223 m^2 (floor area) single-family Canadian home with basement. The wood design is a wood frame structure, the steel design a light-gage steel frame structure and the concrete design is an "insulated concrete forms" structure with composite concrete-steel joist floors. The walls, including the basement walls were adjusted from "normal design" to ensure functional equivalence. The roof structure, cladding and windows was assumed the same for all three alternatives and was excluded from the analysis.

The system boundaries for the LCA are essentially "cradle to gate", i.e. excluding the usage phase, which is assumed equal for all designs, and the recyclable energy from the buildings (recycled *material* is regarded in the material manufacturing phase).

The LCA results for the wood design are: Cradle to gate energy use: 1140 MJ/m^2; GWP (CO_2-equivalents): 280 kg/m^2. For the steel design: Cradle to gate energy use: 1740 MJ/m^2; GWP (CO_2-equivalents): 340 kg/m^2. For the concrete design: Cradle to gate energy use: 2520 MJ/m^2; GWP (CO_2-equivalents): 420 kg/m^2. Recyclable energy and GWP including energy recycling is not available.

1.3 Case 3 and 4: Environmental Impact of a Single Family Building Shell, CORRIM [4]

This report contains two comparisons of two alternative designs of single-family homes in Minneapolis and Atlanta respectively (USA). The two locations are summarised separately below. The alternatives to wood design (the normal practice) are different for the two locations to reflect market condition differences. The system boundaries for the LCA are the same as in case 2.

1.3.1 Case 3: Minneapolis

These houses are 2 storeys plus basement (identical for both designs). The wood design is a typical American wood frame structure and the steel design is a light-gage steel frame structure. The roof structure is a wood truss structure in both designs. No information is given explicitly about the type of insulation but it is stated that the insulation properties are identical.

The LCA results for the wood design are: Cradle to gate energy use: 969 MJ/m^2; GWP (CO_2-equivalents): 207 kg/m^2. For the steel design: Cradle to gate energy use: 1604 MJ/m^2; GWP (CO_2-equivalents): 309 kg/m^2. Recyclable energy and GWP including energy recycling not available.

1.3.2 Case 4: Atlanta

These houses are 1 storey without basement. The structures are identical (wood frame) except the exterior walls which are wood frame walls in the wood design and concrete block walls in the concrete design. No information is given explicitly about the type of insulation but it is stated that the insulation properties are identical.

The LCA results for the wood design are: Cradle to gate energy use: 580 MJ/m^2; GWP (CO_2-equivalents): 100 kg/m^2. For the steel design: Cradle to gate energy use: 810 MJ/m^2; GWP (CO_2-equivalents): 170 kg/m^2. Recyclable energy and GWP including energy recycling not available.

1.4 Case 5: Environmental and Energy Balances of Wood Products and Substitutes, ECE-FAO [5]

This case compares three different alternatives for the construction of a single family home in Germany, a conventional brick house, a wood frame house and a log house (excluded here).

The system boundaries for the LCA are essentially "cradle to grave", but excluding the usage phase, which is assumed equal for all buildings. Recyclable energy from the buldings is accounted for separately (as shown below). The positive effect on GWP of the CO_2-neutral wood residues are deducted from the total GWP CO_2-equivalents. Recycled material is regarded in the material manufacturing phase.

The LCA results for the wood design are: Cradle to gate energy use: 910 MJ/m^2; GWP (CO_2-equivalents) without energy recycling: 660 kg/m^2 and with energy recycling: 580 kg/m^2. For the brick design: Cradle to gate energy use: 1090 MJ/m^2; GWP (CO_2-equivalents) without energy recycling: 840 kg/m^2 and with energy recycling 800 kg/m^2. Recyclable energy not available.

1.5 Case 6: Energy Use and Environmental Impact of New Residential Buildings, Lund Institute of Technology [6]

In this thesis one wood frame multi-family building in Sweden is compared to a re-design of the same building with another structural system. The re-design is a typical concrete frame design with wood frame in-fill exterior walls and non-structural interior walls.

The system boundaries for the LCA are essentially "cradle to grave", including the usage phase. However, since the usage phase is identical in energy terms for the two designs it is excluded in the summary below. Recyclable energy from the buildings is accounted for in both cases (recycled *material* is regarded in the material manufacturing phase). However, in the LCI data regarding energy use for manufacturing of materials, the "feedstock" energy of wood, i.e. the energy content of the wood component, is included as a burden to the material. This is different from the procedure used in all the other studies summarised here. The energy use is therefore re-analysed here in an approximate manner using the LCI value of 1790 MJ/m^3 sawn softwood from ref. [7] (same as cases 1 and 7) instead of the 5508 MJ/m^3 quoted in the thesis.

The LCA results for the wood design are: Cradle to gate energy use: 4540 MJ/m^2; Recyclable energy: 2160 MJ/m^2; Total energy use excluding usage phase: 4540-2160 = 2380 MJ/m^2. The results for the concrete design are: Cradle to gate energy use: 3740 MJ/m^2; Recyclable energy: 1120 MJ/m^2; Total energy use excluding usage phase: 3740-1120 = 2620 MJ/m^2. GWP not available.

The re-calculated approximate results excluding the "feedstock" energy for wood are (not given in [6]): Cradle to gate energy use: 2840 MJ/m^2; Recyclable energy: 2160 MJ/m^2; Total energy use excluding usage phase: 680 MJ/m^2. For the concrete design: Cradle to gate energy use: 3020 MJ/m^2; Recyclable energy: 1120 MJ/m^2; Total energy use excluding usage phase: 1900 MJ/m^2.

1.6 Case 7: LCA of Building Frame Structures – Environmental Impact over the Life Cycle of Wooden and Concrete Frames, Chalmers University of Technology [8]

This report contains a comparison between two different concrete structure designs and a wood frame design for a multi-family complex in Sweden. The concrete alternative with pre-cast elements (hollow core concrete floor elements, pre-cast concrete sandwich wall panels and steel stud drywall partitions) is selected for the summary here. The LCA was not done for a whole building but for an idealised set of building parts representing a typical floor area unit. The building parts were exterior walls, interior walls and floor structure. It should be noted that the noise insulation properties of the floor in the wood design are better than those in this concrete design.

The system boundaries for the LCA are essentially "cradle to grave", including the usage phase (excluded here for same reasons as for case 6). Recyclable energy from the buildings is not accounted for (recycled *material* is regarded in the material manufacturing phase). In the comparison, the "feedstock" energy of the wood components is also included in the report but as a separate entity. In [8], only values for the separate emissions are given. Thus the CO_2 emission values can not be compared to the GWP values from the other studies.

The LCA results for the wood design are: Cradle to gate energy use including feedstock energy:1310 MJ/m^2 and excluding feedstock energy: 840 MJ/m^2; CO_2 emissions: 40 kg/m^2. For the concrete design: Cradle to gate energy use including as well as excluding (marginal difference) feedstock energy:1430 MJ/m^2; CO_2 emissions: 110 kg/m^2. Recyclable energy and GWP not available.

2. Boundary conditions and system definitions in the studies

The scope of the LCA analyses are to some extent different. This applies to the physical boundaries (functional units) of the studied structures as well as to the system boundaries for LCA:s.

The definition of functional unit is described for each study and varies between encompassing the entire building including foundation and basement all the way to cladding and roofing material and, on the other hand, only describing the essential differences for comparable structures. The absolute numbers for energy use and GWP should therefore not be compared directly. Instead the differences are summarised and discussed here. These energy and GWP differences should be reasonably comparable between the different studies if applying the same LCA system boundaries. The primary exception to this is when different surface materials are assumed (to some extent in case 1,5 and 7) or if there is functional differences, e.g. heat and noise insulation (to some extent in case 1 and 7).

The system boundaries for the LCA vary significantly as noted previously, but essentially they can be divided into four categories as in table 1. The service life energy is excluded in all four categories here, which it can be provided that the thermal properties of the compared buildings are identical. Often this is taken as uniform U-values of walls, roofs and floors of the compared buildings. This is the case in all comparisons except case 1 which compares two actual buildings designed to achieve the same overall energy efficiency (which means that the values for this case should have been adjusted slightly in favour of the "concrete design"). The other two important issues are whether or not the inherent energy in the material (primarily in wood), the "feedstock" energy, should be included as an environmental burden or not and whether or not to give credit to the structure for material that can be recycled as energy (again primarily wood). These system boundaries will be discussed further below.

Table 1 The system boundaries for the LCA cases divided into four categories. The different cases in each column are thus reasonably comparable. The results from the different cases are summarised in figures 1 and 2 below.

	A: - Excluding energy recycling - Including wood feedstock energy	B: - Including energy recycling - Including wood feedstock energy	C: - Excluding energy recycling - Excluding wood feedstock energy	D: - Including energy recycling - Excluding wood feedstock energy
1			Case 1a (modified)	Case 1b (original)
2			Case 2 (original)	
3			Case 3 (original)	
4			Case 4 (original)	
5			Case 5a (original)	Case 5b (original)
6		Case 6a (original)	Case 6b (modified)	Case 6c (modified)
7	Case 7a (original)		Case 7b (modified)	

3. Results from comparisons

The results from the different cases are summarised in figures 1 and 2. The overall conclusion that can be drawn directly from these figures is that in all cases, the wood structure results in lower energy use and GWP than the alternatives, regardless of system boundary conditions applied in the different studies. The next question is then how big this difference is and how significant it is in comparison to e.g. service life energy consumption and GWP type emissions. Figure 2 would imply that the GWP difference in CO_2-equivalents would be somewhere in the range of 50-200 kg/m^2 for LCA system boundary category C according to table 1 and 200-300 kg/m^2 for category D (adjusting the values from case 1b somewhat to account for the deviations mentioned above).

In [7], four different multi-family buildings in the southern half of Sweden are studied to determine their energy need also during occupancy. The calculated total energy need (heating, ventilation, hot water and electricity) varies between 100 and 150 kWh/m^2 (360-540 MJ/m^2). Based on the normal energy mix for producing district heating, which is the normal energy supply for multi-family dwellings in Sweden, the total GWP is thus estimated as 20-26 kg/m^2 CO2-equivalent emissions per year. This means that the effect of choosing wood construction instead of the alternatives can be comparable to the occupation phase GWP during 10-15 years (of a normally assumed economic service life of around 50 years), assuming category D LCA system boundaries. If comparing with the average energy source for all housing in Sweden the effect is even higher since district heating (with relatively low GWP) is not normally supplied for single-family housing.

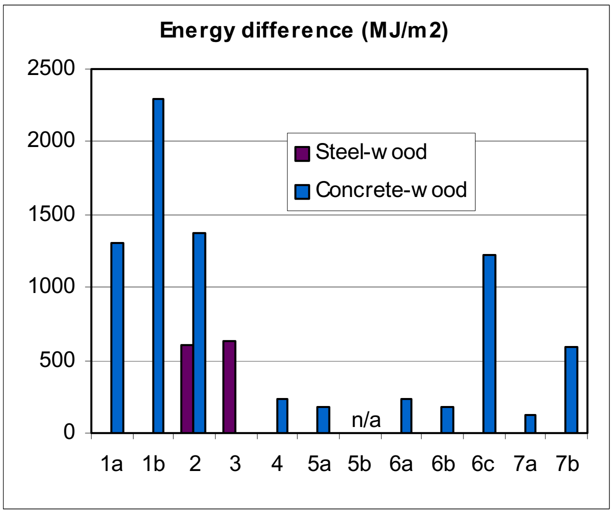

Fig. 1 Energy usage differences between the wood versus the concrete and steel building alternatives respectively compared in the different studies. Note that the system boundaries (and to some extent the physical boundaries) vary according to table 1.

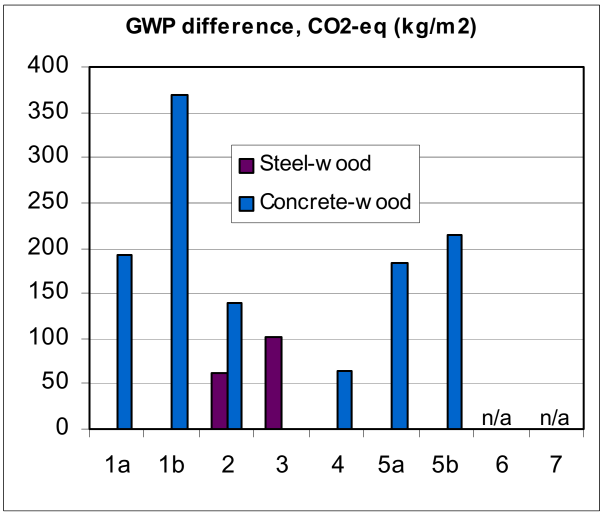

Fig. 2 Global warming potential (GWP) differences between the compared buildings in the different studies, given as CO2-equivalents. Note that the system boundaries (and to some extent the functional units) vary according to table 1.

4. Discussion of LCA system boundaries

As described above, the selection of system boundaries for the LCA has large effects on the results of a comparative LCA. Indeed this is one of the reasons for a general feeling of frustration towards the LCA tool for environmental impact assessment; More or less anything can be proven depending on the system definitions!

The special issue when comparing wood to steel and concrete is that wood is a renewable material with an inherent energy potential whereas the others are finite (mineral) resources with no energy potential. Plastic materials share the latter part with wood, i.e. they have an inherent energy, but are primarily extracted from finite (fossil) resources.

For steel and concrete, the material recycling is normally taken into account to the extent corresponding to current recycling practices. This is treated as a partial input to the production process of new material and as a decrease of the volume that goes to landfill. For wood, normally the material reuse or recycling share is lower than for e.g. steel. Instead the normal practice is to use it for energy recycling purposes, which is quite natural for a renewable resource if this is a more economical end use than material recycling.

In LCA terminology this field is denoted system expansion. Let's examine the four categories A to D of system boundaries listed in table 1. A) can be disregarded immediately since it is clearly not logical to burden the wood structure with its' inherent energy and then not credit it with its' energy recycling potential. B) could be a feasible alternative if the raw material could just as well be utilised as a renewable energy resource as for wood products. However, economically this is not the case because the resource extraction (forest harvesting) could not be carried out purely for energy production purposes since the raw material price would then not cover the harvesting and certainly not a sustainable forest management at today's level. C) would only be logical if the wood material is not likely to be energy recycled after its' use. D) therefore seems to be the appropriate category since the use of logs as wood products is required economically to be able to extract the raw material and when finally the wood product is available at the end of its service life it provides a CO_2-neutral energy resource. This is of course provided that the raw material comes from a forestry that is truly sustainable.

What then needs to be dealt with is how to credit the wood building with the recycled energy. This seems to be done in two different ways in the different studies. In case 1, the recycling energy is treated as a resource that will replace another type of additional energy production. In Sweden today, this would be energy produced from fossil fuels (oil or coal). The amount of GWP from this energy production is then added to the alternative building structure as a burden (alternatively it could have been subtracted from the GWP of the wood alternative, giving a negative GWP). In case 5, the difference between the sub-cases 5a and 5b is surprisingly small. This may imply that the crediting is done in another way or that the energy recycling is erroneously calculated (not given explicitly in [5]). An important issue to discuss is of course what type of energy the recycled wood product is likely to eventually replace since this is probably 50 or more years into the future.

5. Conclusions

An increased use of wood in construction will undoubtedly have a positive effect on the total energy use and emissions of greenhouse gases.

In a European scale, the GWP decrease potential from housing construction can be roughly estimated. Each year approximately 1,8 million housing units are built within the EU 15 area. An estimated 5 percent or less of this has wood structure, whereas the rest is totally dominated by concrete structures and mineral based block or brick structures. The average living area is presently around 100 m^2/unit in new construction. If all housing was built with wood structures instead this would mean a further 1,7 million units at 100 m^2/unit times 200-300 kg/m^2 less GWP. This sums up to 35-50 Mton CO_2 equivalents per year from the substitution effect only, which is equal to 0,9-1,3 % of the total annual EU 15 emissions (3900 Mton) or 11-16 % of the decrease commitment according to the Kyoto protocol (8% decrease of the 1990 figures). This would require roughly a further 35 million m^3 of sawn softwood (compared to today's consumption of roughly 100 million m^3), which is probably possible within the current European forest practice.

It should also be evident from this report that the building material sector and its' researchers need a better agreement around the methodologies as well as tools and data for LCA of building structures.

6. Acknowledgements

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7. References

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